

Data integrity Report

Missing Values

What I found? :

- The dataset contained missing values (NaN entries) across multiple columns rather than being isolated to a single attribute. These null values resulted from different data sources and operational scenarios

Where was it? :

- **checkout_id**: Missing due to records scraped from the web-based *Reading Kickoff* event, where transactions were logged without database primary key IDs.
- **return_date** : Missing because certain books are currently checked out or were borrowed during event checkouts where no return date was initially recorded.
- **publication_year** : Omitted or incomplete entries within the library catalog database.
- **grade** : Unspecified student grade levels in member registration forms.
- **join_date** & Member Demographics (**first_name**, **last_name**, **neighborhood**, **membership_status**): Missing due to incomplete registration profiles or foreign key mismatches with the members table.

How big was it ? :

checkout_id (26 missing values)
return_date (91 missing values)
publication_year (35 missing values)
grade (41 missing values)
join_date (11 missing values)
Member Demographics (**first_name**, **last_name**, **neighborhood**, **membership_status**) - (5 missing values each)

What did I do and why ? :

checkout_id :

- Action: Filled missing entries with the string "**WEB_EVENT**".
- Reason: These transactions originated from the web-scraped *Reading Kickoff* event and lacked database primary keys; assigning a static indicator preserves real checkout records without inventing fake numerical IDs.

return_date :

- Action: Filled missing entries with the string "**Not Returned Yet**".
- Reason: Missing dates represent active, ongoing book loans or event borrowings; deleting these rows would undercount actual library circulation.

publication_year :

- Action: Imputed missing values using the statistical Mode of publication years.
- Reason: Preserves the numeric data type and prevents artificial skewing without discarding valid borrowing transactions.

grade :

- Action: Imputed missing values using the most frequent academic grade level (Mode).
- Reason: Ensures complete student demographic records while maintaining the overall grade distribution.

Missing Personal Details (`first_name`, `last_name`, `neighborhood`, `membership_status`):

- **Action:** Resolved automatically by filtering out invalid member records in Problem 4.
- **Reason:** Missing personal details were a direct artifact of invalid foreign keys (unregistered `member_ids` not present in the main database).

`Join_date` :

- **Action:** Filled missing entries for registered members with the explicit string "`Unknown`", while the remaining cases were resolved automatically by filtering out invalid member records in Problem 4.
- **Reason:** Missing join dates occurred either due to incomplete member profile logs or orphan transactions linked to ghost member IDs; using an explicit placeholder preserves transaction validity for registered users without inventing historical membership timestamps.

Duplicates

What did I find ? :

The dataset contained duplicate rows that made the total number of book checkouts look higher than it actually was.

When I checked the data, I found duplicates where every single column matched completely. I made sure to only count these as duplicates, while keeping real repeat checkouts (where a student borrows the same book on a different date).

Where was it ? :

Showed up across all 14 columns as full-row identical duplicates (repeating exact matches of **checkout_id**, **member_id**, **book_id**, and **checkout_date**).

How big it was :

Number of extra records: 8 duplicate records

Total records involved: 16 total rows

(8 original transactions + 8 exact duplicate copies).

Percentage of dataset affected:

~3.8% of the dataset (16 out of 423 rows).

What did I do and why ? :

The Decision I Made:

I decided to delete 100% exact full-row duplicates (8 duplicated rows) while keeping similar records where a student borrowed the same book on different dates.

Why:

Leaving exact duplicate records in the dataset artificially inflates library circulation numbers and distorts student borrowing statistics. Removing them was necessary to restore accurate transaction counts.

Inconsistent format

What did i find ? :

Categorical text fields across the dataset contained structural noise, non-standard formatting, and leading/trailing whitespace artifacts.

Where it was ? :

Primary Categorical Columns:

- **neighborhood**: Contained leading and trailing whitespace artifacts (e.g., " **Zamalek** " vs "**Zamalek**").
- **membership_status**: Contained inconsistent letter capitalization (e.g., mixing '**active**', '**Active**', and '**ACTIVE**').

Secondary Text Columns:

- **title & genre**: Contained minor whitespace variations and inconsistent letter casing across book titles and genre categories.

How big it was? :

The Problem Size:

The issue wasn't broken data, but extra spaces (like " **Zamalek** ") and lowercase letters (like '**active**' instead of '**Active**') scattered across columns like **neighborhood**, **membership_status**, **genre**, and **title**.

Even though the data was there, these small formatting differences would trick Python or SQL into treating '**Active**' and '**active**' as two completely different categories.

What did I do and why ?:

I cleaned up the text columns (**neighborhood**, **membership_status**, **title**, and **genre**) by stripping out all extra spaces at the beginning and end using **.str.strip()**, and then standardized the capitalization to Title Case using **.str.title()**.

Why:

To make sure every category is written in the exact same way across the whole dataset. This prevents duplicate categories (like **"Zamalek "** and **"Zamalek"**, or **'active'** and **'Active'**) from messing up our groups, filters, and charts when we analyze the data

Checkouts with no matching members

What i found ?:

There were checkout records linked to member IDs that don't exist in the official members database.

Because of this mismatch, all their personal info (**names, neighborhoods, join dates**) appeared as missing

Where it was ?:

The issue showed up specifically in the **member_id** column.

How big it was? :

it appeared in 5 checkout records that contained invalid member IDs (**1104, 1201, and 1150**) which didn't match any registered user in the main members database.

What did I do ? and Why ? :

I filtered out and removed the 5 checkout records associated with the non-existent member IDs during data cleaning

Why I Removed Them

- **To Keep Database Rules:** In databases, you can't have a transaction tied to a member that doesn't actually exist.
- **To Protect Our Analytics:** Since these 5 rows had no profile info (no names, grades, or neighborhoods), keeping them would mess up our student and neighborhood stats in Task 3. Removing them ensures we only analyze real, registered members.

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